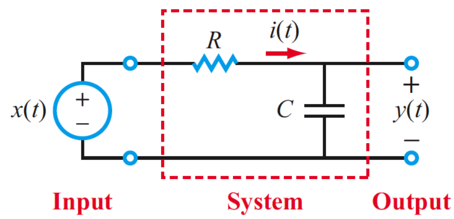
ECE 3337: Signals & Systems Analysis,
Class Worksheet – Lecture 4

1. (15 pts) Are the following systems *i*) linear and/or *ii*) time-invariant, and why?
   a. (5 pts) $y(t) = ax(t) + b$
   b. (5 pts) $y(t) = atx(t)$
   c. (5 pts) $\frac{dy}{dt} + ay(t) = \frac{dx}{dt}$

2. (20 pts) By systematically testing for each of the relevant properties, prove that the RC circuit shown represents a linear time-invariant system that is also causal.



(a) RC circuit

3. (45 pts) Calculate the output of the above RC circuit for the input $x(t) = au(t) + b\delta(t)$
   a. (5 pts) Derive the differential equation using circuit theory
   b. (10 pts) Perform a direct solution of the differential equation from part a.
   c. (5 pts) Explicitly confirm that your answer to part b. is causal
   d. (5 pts) Write down the text-book provided formulae for the impulse response h(t) and the step response $y_{step}(t)$ for this circuit
   e. (10 pts) Using superposition, calculate the output of the system from part d.
   f. (5 pts) Explicitly verify that your answer in part e. is causal
   g. (5 pts) Confirm that your answer to parts b. and e. are identical.

4. (20 pts, 5 point trying, 5 each ,a,b,c) Calculate the response of the RC circuit with RC=1 to a pulse with an amplitude of 1V and duration of 2 seconds.
   a. (5 pts) Write down the formula for the input and output voltages
   b. (5 pts) Calculate the peak output voltage, and the time when it occurs
   c. (5 pts) What happens to the current i(t) when the output voltage is maximum?